

# Matrix Theory: Lecture 4

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# From Last Lecture

Image and kernel (subspaces):

$$\operatorname{Im} \mathcal{A} = \{\mathcal{A}x \mid x \in V\}$$

$$\operatorname{Ker} \mathcal{A} = \{v \in V \mid \mathcal{A}v = 0\}$$

## Theorem 5.12

Let  $\mathcal{A} : V \rightarrow U$  be a linear map. Then

$$\dim \operatorname{Im} \mathcal{A} + \dim \operatorname{Ker} \mathcal{A} = \dim V.$$

If  $A$  is a  $m \times n$  matrix then

$$r(A) + \dim \operatorname{Ker} A = n.$$



# Outline

- Diagonalization
- Spectral theory
- The Jordan Normal form
- Jordan form for Nilpotent operators



# Diagonalization

Diagonalization simplifies problems by disentangling equations:

Ex.

$$\begin{cases} x_1'(t) &= -x_1(t) + 2x_2(t) \\ x_2'(t) &= -3x_1(t) + 4x_2(t) \end{cases} \Leftrightarrow \underbrace{\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix}}_{x'} = \underbrace{\begin{pmatrix} -1 & 2 \\ -3 & 4 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}}_x.$$

With  $S = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$  we get

$$A = \begin{pmatrix} -1 & 2 \\ -3 & 4 \end{pmatrix} = SDS^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}.$$

Switch coordinates  $y = Sx$

$$\underbrace{\begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix}}_{y'} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}}_D \underbrace{\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}}_y \Leftrightarrow \begin{cases} y_1'(t) &= y_1(t) \\ y_2'(t) &= 2y_2(t) \end{cases}.$$

Solve

$$\begin{cases} y_1(t) &= C_1 e^t \\ y_2(t) &= C_2 e^{2t} \end{cases} \quad \text{and } x = S^{-1}y...$$



# Diagonalization, Eigenvalues and Eigenvectors

Def. A vector  $v \neq 0$  is called an **eigenvector** for a linear operator  $\mathcal{A}$  if

$$\mathcal{A}v = \lambda v$$

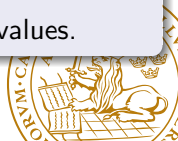
for some scalar  $\lambda$ , which is called an **eigenvalue** for  $\mathcal{A}$  (and  $v$ ).  
(We make a similar definition for matrices.)

## Theorem 6.5

Eigenvectors  $X_1, \dots, X_k$  (for a matrix  $A$ ) with different eigenvalues  $\lambda_1, \dots, \lambda_k$  are linearly independent.

## Theorem 6.6

- 1  $A$  is diagonalizable  $\Leftrightarrow A$  has  $n$  linearly independent eigenvectors.
- 2  $A$  is diagonalizable if it has  $n$  distinct (all different) eigenvalues.



# The Characteristic Polynomial

## Theorem 6.7

$\lambda$  is an eigenvalue of  $A$  if and only if it fulfills the **characteristic equation**

$$p_A(\lambda) = 0,$$

where  $p_A(\lambda) = |\lambda I - A|$  is the **characteristic polynomial**.

## Corollary 6.8

If  $A$  is triangular then  $\lambda_i = a_{ii}$ .

## Corollary 6.9

Two matrices that are similar have the same characteristic polynomial and eigenvalues.

Remark: Not sufficient to test if two matrices are similar.



# Jordanization

Diagonalization may not be necessary.

Ex.

$$\begin{cases} x_1'(t) = x_1(t) + x_2(t) \\ x_2'(t) = -x_1(t) + 3x_2(t) \end{cases} \Leftrightarrow \underbrace{\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix}}_{x'} = \underbrace{\begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}}_x.$$

With  $S = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$  we get

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix} = S J_2(2) S^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}.$$

Switch coordinates  $y = Sx$

$$\underbrace{\begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix}}_{y'} = \underbrace{\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}}_{J_2(2)} \underbrace{\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}}_y \Leftrightarrow \begin{cases} y_1'(t) = 2y_1(t) + y_2(t) \\ y_2'(t) = 2y_2(t) \end{cases}.$$

Solve

$$\begin{cases} y_1(t) = C_1 e^{2t} + C_2 t e^{2t} \\ y_2(t) = C_2 e^{2t} \end{cases} \quad \text{and } x = S^{-1}y...$$



# The General Jordan Normal Form

## Theorem 7.8

For every operator  $\mathcal{A}$  there exists a basis consisting of chains of form

$$\begin{array}{ccccccc} (\mathcal{A} - \lambda_1 I)^{k_1-1} v_1, & (\mathcal{A} - \lambda_1 I)^{k_1-2} v_1, & \cdots, & (\mathcal{A} - \lambda_1 I) v_1, & v_1 \\ (\mathcal{A} - \lambda_2 I)^{k_2-1} v_2, & (\mathcal{A} - \lambda_2 I)^{k_2-2} v_2, & \cdots, & (\mathcal{A} - \lambda_2 I) v_2, & v_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{array}$$

with the conditions  $(\mathcal{A} - \lambda_i I)^{k_i} v_i = 0$ . All  $\lambda_i$  are eigenvalues and some of them may be equal. Relative to this basis the operator has matrix

$$J = \begin{pmatrix} J_{k_1}(\lambda_1) & O & \cdots & O \\ O & J_{k_2}(\lambda_2) & \cdots & O \\ \cdots & \cdots & \cdots & \cdots \\ O & O & \cdots & J_{k_m}(\lambda_m) \end{pmatrix},$$

which is called the **Jordan form** of the operator.



# The Jordan Normal Form (for Nilpotents)

Ex1.  $A = \begin{pmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$  nilpotent since  $A^3 = 0$ , eigenvalues 0, not

diagonalizable. Similar to  $J_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$  since

$$\underbrace{\begin{pmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 2 & 2 & -3 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}}_S = \begin{pmatrix} 0 & 2 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & 2 & -3 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}}_S \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}}_{J_3}$$

Columns of  $S$  form a chain (of basis elements):

$$A \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$



# The Jordan Normal Form (for Nilpotents)

## Theorem 7.3

If  $\mathcal{A}^k v = 0$  but  $\mathcal{A}^{k-1} v \neq 0$  then the vectors in the **chain**

$$\mathcal{A}^{k-1} v, \mathcal{A}^{k-2} v, \dots, \mathcal{A} v, v$$

are linearly independent.



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$$\mathcal{A}^{k-1} v, \mathcal{A}^{k-2} v, \dots, \mathcal{A} v, v$$

are linearly independent.

$$\alpha_{k-1} \mathcal{A}^{k-1} v + \alpha_{k-2} \mathcal{A}^{k-2} v + \dots + \alpha_1 \mathcal{A} v + \alpha_0 v = 0$$



# The Jordan Normal Form (for Nilpotents)

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If  $\mathcal{A}^k v = 0$  but  $\mathcal{A}^{k-1} v \neq 0$  then the vectors in the **chain**

$$\mathcal{A}^{k-1} v, \mathcal{A}^{k-2} v, \dots, \mathcal{A} v, v$$

are linearly independent.

$$\mathcal{A}(\alpha_{k-1} \mathcal{A}^{k-1} v + \alpha_{k-2} \mathcal{A}^{k-2} v + \dots + \alpha_1 \mathcal{A} v + \alpha_0 v) = 0$$



# The Jordan Normal Form (for Nilpotents)

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If  $\mathcal{A}^k v = 0$  but  $\mathcal{A}^{k-1} v \neq 0$  then the vectors in the **chain**

$$\mathcal{A}^{k-1} v, \mathcal{A}^{k-2} v, \dots, \mathcal{A} v, v$$

are linearly independent.

$$\alpha_{k-1} \underbrace{\mathcal{A}^k v}_{=0} + \alpha_{k-2} \mathcal{A}^{k-1} v + \dots + \alpha_1 \mathcal{A}^2 v + \alpha_0 \mathcal{A} v = 0$$



# The Jordan Normal Form (for Nilpotents)

## Theorem 7.3

If  $\mathcal{A}^k v = 0$  but  $\mathcal{A}^{k-1} v \neq 0$  then the vectors in the **chain**

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are linearly independent.

$$\alpha_{k-2} \mathcal{A}^{k-1} v + \dots + \alpha_1 \mathcal{A}^2 v + \alpha_0 \mathcal{A} v = 0$$



# The Jordan Normal Form (for Nilpotents)

## Theorem 7.3

If  $\mathcal{A}^k v = 0$  but  $\mathcal{A}^{k-1} v \neq 0$  then the vectors in the **chain**

$$\mathcal{A}^{k-1} v, \mathcal{A}^{k-2} v, \dots, \mathcal{A} v, v$$

are linearly independent.

$$\alpha_{k-2} \mathcal{A}^{k-1} v + \dots + \alpha_1 \mathcal{A}^2 v + \alpha_0 \mathcal{A} v = 0$$

Repeat until one term with non-zero  $\alpha_i$  remains:

$$\underbrace{\alpha_i}_{\neq 0} \mathcal{A}^{k-1} v = 0 \Rightarrow \mathcal{A}^{k-1} v = 0 \quad \text{Contradiction!}$$



# The Jordan Normal Form (for Nilpotents)

Ex3.

$$\underbrace{\begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 2 & 2 & -3 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}}_S = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} =$$

$$\underbrace{\begin{pmatrix} 2 & 2 & -3 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}}_S \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_J = S \left( \begin{array}{c|c} J_2(0) & 0 \\ \hline 0 & J_1(0) \end{array} \right)$$

$$A \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

One chain of length 2 and one of length 1.





# The Jordan Normal Form (for Nilpotents)

$$\text{Ex3.} \quad \underbrace{\begin{pmatrix} 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_S = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} =$$

$$\underbrace{\begin{pmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_S \underbrace{\begin{pmatrix} 0 & 1 & | & 0 & 0 \\ 0 & 0 & | & 0 & 0 \\ \hline 0 & 0 & | & 0 & 1 \\ 0 & 0 & | & 0 & 0 \end{pmatrix}}_J = S \left( \begin{array}{c|c} J_2(0) & 0 \\ \hline 0 & J_2(0) \end{array} \right)$$

$$A \begin{pmatrix} 3 \\ 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Two chains of length 2.



# The Jordan Normal Form (for Nilpotents)

## Lemma 7.4

Suppose that we have  $s$  chains of the form

$$\mathcal{A}^{k_i-1}v_i, \mathcal{A}^{k_i-2}v_i, \dots, \mathcal{A}v_i, v_i$$

with  $\mathcal{A}^{k_i}v_i = 0$ . If the vectors  $\mathcal{A}^{k_1-1}v_1, \mathcal{A}^{k_2-1}v_2, \dots, \mathcal{A}^{k_s-1}v_s$  (the last ones of every chain) are independent then all vectors of all chains are independent.



# The Jordan Normal Form (for Nilpotents)

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$$\begin{aligned} & \alpha_{k_1-1}\mathcal{A}^{k_1-1}v_1 + \alpha_{k_1-2}\mathcal{A}^{k_1-2}v_1 + \dots + \alpha_1\mathcal{A}v_1 + \alpha_0v_1 & + \\ & \beta_{k_2-1}\mathcal{A}^{k_2-1}v_2 + \beta_{k_2-2}\mathcal{A}^{k_2-2}v_2 + \dots + \beta_1\mathcal{A}v_2 + \beta_0v_2 & + \\ & \dots & \\ & \gamma_{k_s-1}\mathcal{A}^{k_s-1}v_s + \gamma_{k_s-2}\mathcal{A}^{k_s-2}v_s + \dots + \gamma_1\mathcal{A}v_s + \gamma_0v_s & \\ & = 0 \end{aligned}$$



# The Jordan Normal Form (for Nilpotents)

## Lemma 7.4

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with  $\mathcal{A}^{k_i}v_i = 0$ . If the vectors  $\mathcal{A}^{k_1-1}v_1, \mathcal{A}^{k_2-1}v_2, \dots, \mathcal{A}^{k_s-1}v_s$  (the last ones of every chain) are independent then all vectors of all chains are independent.

Multiplication with  $\mathcal{A}$ :

$$\begin{aligned} & \alpha_{k_1-1}\mathcal{A}^{k_1}v_1 + \alpha_{k_1-2}\mathcal{A}^{k_1-1}v_1 + \dots + \alpha_1\mathcal{A}^2v_1 + \alpha_0\mathcal{A}v_1 & + \\ & \beta_{k_2-1}\mathcal{A}^{k_2}v_2 + \beta_{k_2-2}\mathcal{A}^{k_2-1}v_2 + \dots + \beta_1\mathcal{A}^2v_2 + \beta_0\mathcal{A}v_2 & + \\ & \dots & \\ & \gamma_{k_s-1}\mathcal{A}^{k_s}v_s + \gamma_{k_s-2}\mathcal{A}^{k_s-1}v_s + \dots + \gamma_1\mathcal{A}^2v_s + \gamma_0\mathcal{A}v_s & \\ & = 0 \end{aligned}$$



# The Jordan Normal Form (for Nilpotents)

## Lemma 7.4

Suppose that we have  $s$  chains of the form

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First column vanishes:

$$\begin{aligned} & \alpha_{k_1-2}\mathcal{A}^{k_1-1}v_1 + \dots + \alpha_1\mathcal{A}^2v_1 + \alpha_0\mathcal{A}v_1 & + \\ & \beta_{k_2-2}\mathcal{A}^{k_2-1}v_2 + \dots + \beta_1\mathcal{A}^2v_2 + \beta_0\mathcal{A}v_2 & + \\ & \dots & + \\ & \gamma_{k_s-2}\mathcal{A}^{k_s-1}v_s + \dots + \gamma_1\mathcal{A}^2v_s + \gamma_0\mathcal{A}v_s & \\ & = 0 \end{aligned}$$



# The Jordan Normal Form (for Nilpotents)

## Lemma 7.4

Suppose that we have  $s$  chains of the form

$$\mathcal{A}^{k_i-1}v_i, \mathcal{A}^{k_i-2}v_i, \dots, \mathcal{A}v_i, v_i$$

with  $\mathcal{A}^{k_i}v_i = 0$ . If the vectors  $\mathcal{A}^{k_1-1}v_1, \mathcal{A}^{k_2-1}v_2, \dots, \mathcal{A}^{k_s-1}v_s$  (the last ones of every chain) are independent then all vectors of all chains are independent.

Repeat until only one column with non-zero elements remains:

$$\begin{aligned} & \alpha_1 \mathcal{A}^{k_1-1}v_1 && + \\ & \beta_1 \mathcal{A}^{k_2-1}v_2 && + \\ & \dots && + \\ & \gamma_1 \mathcal{A}^{k_s-1}v_s && \\ & = 0 && \text{Contradicting independence!} \end{aligned}$$



# The Jordan Normal Form (for Nilpotents)

## Theorem 7.5

For every nilpotent operator  $\mathcal{A} : V \mapsto V$  there exists a basis of  $V$  consisting of  $s$  chains of length  $k_i$ ,  $i = 1, \dots, s$ , with  $\mathcal{A}^{k_i} v_i = 0$ . In this basis the operator  $\mathcal{A}$  has the matrix

$$J = \begin{pmatrix} J_{k_1} & 0 & \dots & 0 \\ 0 & J_{k_2} & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & J_{k_s} \end{pmatrix},$$

where  $J_k$  is a Jordan block of size  $k \times k$  of the type

$$J_k = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

# The Jordan Normal Form (for Nilpotents)

Proof outline: Induction over  $\dim V$ .

- 1  $\dim V = 1$ . One vector basis  $v$  and  $\mathcal{A}v = 0$ .





# The Jordan Normal Form (for Nilpotents)

Proof outline: Induction over  $\dim V$ .

- ②  $\dim V = n$ . Assume true for  $\dim V \leq n - 1$ .

Look at  $U := \operatorname{Im} \mathcal{A}$ .

$$\dim U = \dim(\operatorname{Im} \mathcal{A}) = \dim V - \underbrace{\dim(\operatorname{Ker} \mathcal{A})}_{\geq 1}$$

Gives chain-basis for  $\operatorname{Im} \mathcal{A}$ :

$$\begin{array}{ccccccc} \mathcal{A}^{l_1-1}u_1, & \mathcal{A}^{l_1-2}u_1, & \dots & \mathcal{A}u_1, & u_1 & & \\ \mathcal{A}^{l_2-1}u_2, & \mathcal{A}^{l_2-2}u_2, & \dots & \mathcal{A}u_2, & u_2 & & \\ \dots & \dots & \dots & \dots & \dots & \dots & \\ \mathcal{A}^{l_p-1}u_p, & \mathcal{A}^{l_p-2}u_p, & \dots & \mathcal{A}u_p, & u_p & & \end{array}$$



# The Jordan Normal Form (for Nilpotents)

Proof outline: Induction over  $\dim V$ .

③ Extend with elements  $v_i$  such that  $u_i = \mathcal{A}(v_i)$ :

$$\begin{array}{ccccccc} \mathcal{A}^{l_1-1}u_1, & \mathcal{A}^{l_1-2}u_1, & \cdots & \mathcal{A}u_1, & u_1 & v_1 \\ \mathcal{A}^{l_2-1}u_2, & \mathcal{A}^{l_2-2}u_2, & \cdots & \mathcal{A}u_2, & u_2 & v_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \mathcal{A}^{l_p-1}u_p, & \mathcal{A}^{l_p-2}u_p, & \cdots & \mathcal{A}u_p, & u_p & v_p \end{array}$$



# The Jordan Normal Form (for Nilpotents)

Proof outline: Induction over  $\dim V$ .

- ③ Extend with elements  $v_i$  such that  $u_i = \mathcal{A}(v_i)$ :

$$\begin{array}{ccccccc} \mathcal{A}^{l_1} v_1, & \mathcal{A}^{l_1-1} v_1, & \dots & \mathcal{A}^2 v_1, & \mathcal{A} v_1 & v_1 \\ \mathcal{A}^{l_2} v_2, & \mathcal{A}^{l_2-1} v_2, & \dots & \mathcal{A}^2 v_2, & \mathcal{A} v_2 & v_2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \mathcal{A}^{l_p} v_p, & \mathcal{A}^{l_p-1} v_p, & \dots & \mathcal{A}^2 v_p, & \mathcal{A} v_p & v_p \end{array}$$



# The Jordan Normal Form (for Nilpotents)

Proof outline: Induction over  $\dim V$ .

④ Extend with elements from  $\text{Ker } \mathcal{A}$ :

$$\begin{array}{ccccccc}
 \mathcal{A}^{l_1} v_1, & \mathcal{A}^{l_1-1} v_1, & \dots & \mathcal{A}^2 v_1, & \mathcal{A} v_1 & v_1 \\
 \mathcal{A}^{l_2} v_2, & \mathcal{A}^{l_2-1} v_2, & \dots & \mathcal{A}^2 v_2, & \mathcal{A} v_2 & v_2 \\
 \dots & \dots & \dots & \dots & \dots & \dots \\
 \mathcal{A}^{l_p} v_p, & \mathcal{A}^{l_p-1} v_p, & \dots & \mathcal{A}^2 v_p, & \mathcal{A} v_p & v_p \\
 & & & & & v_{p+1} \\
 & & & & & v_{p+2} \\
 & & & & & \dots \\
 & & & & & v_s
 \end{array}$$

$s = \dim \text{Ker } \mathcal{A}$ .

⑤ Show that this is a basis.

